Tour de Maths 2024

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1. The cost of a flight to Cape Town including 15% VAT is R1403. What is the cost excluding VAT?

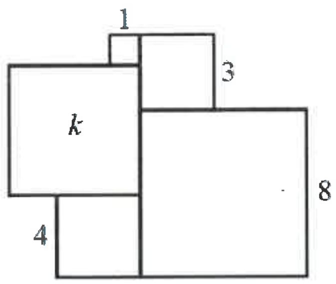
2. Hagrid has 100 animals. Among these animals,
   - each is either striped or spotted but not both,
   - each has either wings or horns but not both,
   - there are 28 striped animals with wings,
   - there are 62 spotted animals, and
   - there are 36 animals with horns.

   How many of Hagrid's spotted animals have horns?

3. If $10^x \times 10^y \times 10^z = 10^6$ then what is the average of x, y and z?

4. What is the unit digit of: $5^{2019} - 3^{2019}$?

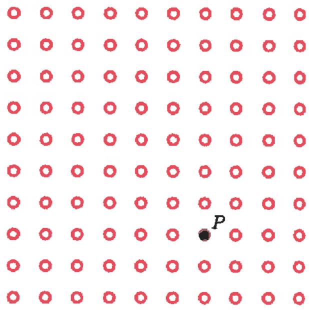
5. In the diagram, the side lengths of four squares are shown. The area of the fifth square is *k*. What is the value of *k*?



6. The positive integers from 1 to 576 are written in a 24·24 grid so that the first row contains the numbers 1 to 24, the second row contains the numbers 25 to 48, and so on, as shown. An 8·8 square is drawn around 64 of these numbers. (These 64 numbers consist of 8 numbers each in 8 rows.) The sum of the numbers in the four corners of the 8·8 square is 1646. What is the number in the bottom right corner of this 8·8 square?

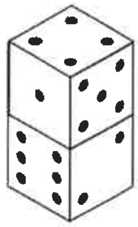
1	2	$\cdots$	23	24
25	26	$\cdots$	47	48
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
553	554	$\cdots$	575	576

7. A 10 by 10 grid is created using 100 small circles, as shown. The circle labelled P has been coloured in on the grid. One of the other 99 circles on the grid is randomly chosen to be labelled Q and coloured in as well. What is the probability the line segment connecting P and Q is vertical or horizontal?



8. Pat rides her bike at 20 km/h and Chris ride his bike at 15 km/h. At 12 noon, Chris is 2 km east of Pat, and each begins to ride east. At what time will Pat catch Chris?

9. Two cubes are stacked as shown. The faces of each cube are labelled with 1, 2, 3, 4, 5, and 6 dots. A total of five faces are shown. What is the total number of dots on the other seven faces of these two cubes?



10. Four matchsticks are used to construct the first figure, 10 for the second, 18 for the third and so on. How many matchsticks are needed to make the 30th figure?

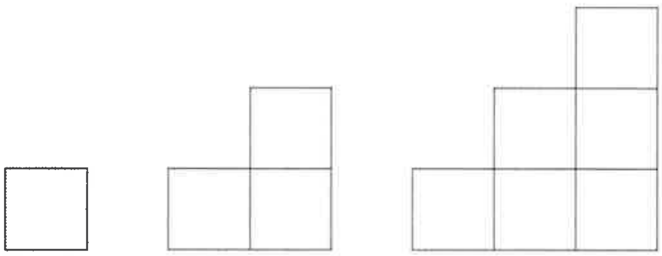
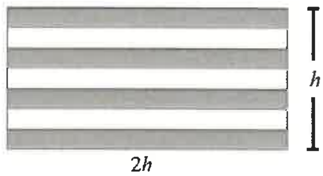


Figure 1 Figure 2 Figure 3

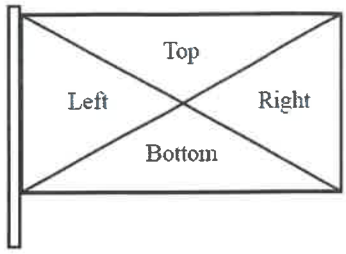
11. The rectangular flag shown is divided into seven stripes of equal height. The height of the flag is h and the length of the flag is twice its height. The total area of the four shaded regions is 1 400 cm^2. What is the height of the flag?



12. In the 4×4 grid shown, each of the four symbols has a different value. The sum of the values of the symbols in each row is given to the right of that row. What is the value of ϕ?

♡	△	△	♡	26
△	△	△	△	24
□	♦	♡	♦	27
□	♡	□	△	33

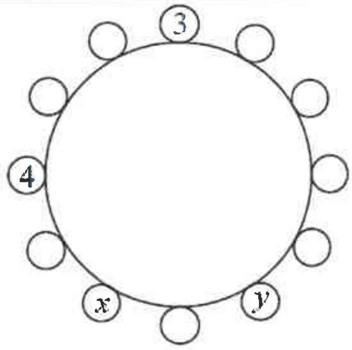
13. A rectangular flag is divided into four triangles, labelled Left, Right, Top, and Bottom, as shown. Each triangle is to be coloured one of red, white, blue, green, and purple so that no two triangles that share an edge are the same colour. How many different flags can be made?



14. In a magic square, the numbers in each row, the numbers in each column, and the numbers on each diagonal have the same sum. In the magic square shown, the sum a + b + c equals?

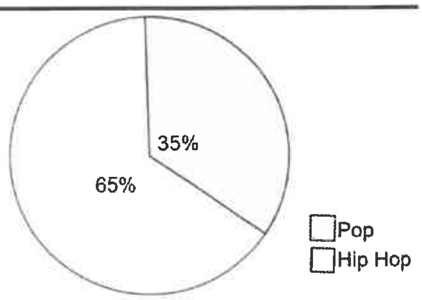
a	13	b
19	c	11
12	d	16

15. Each integer from 1 to 12 is to be placed around the outside of a circle so that the positive difference between any two integers next to each other is at most 2. The integers 3, 4, x, and y are placed as shown. What is the value of x + y?



16. The sum of two numbers is 384 and their HCF is 48. What is the largest difference between the two numbers?

17. The graph shows styles of music on a playlist. Country music songs are added to the playlist so that now 40% of the songs are Country. If the ratio of Hip Hop songs to Pop songs remains the same, what percentage of the total number of songs are now Hip Hop?



18. ABC is a three-digit integer where the first digit is A, second digit is B and third digit is C. Similarly, DEF is a three-digit integer where the first digit is D, second digit is E and third digit is F. We are given that:

$$\begin{array}{r} ABC \\ +DEF \\ \hline 1234 \end{array}$$

What is the largest possible value of A + B + C + D + E + F?

19. There are n students in the Mathematics club at Scoins College. When Mrs. Fryer tries to put the n students in groups of 4, there is one group with fewer than 4 students, but all of the other groups are complete. When she tries to put the n students in groups of 3, there are 3 more complete groups than there were with groups of 4, and there is again exactly one group that is not complete. When she tries to put the n students in groups of 2, there are 5 more complete groups than there were with groups of 3, and there is again exactly one group that is not complete.

What is the sum of the digits of: $n^2 - n$?

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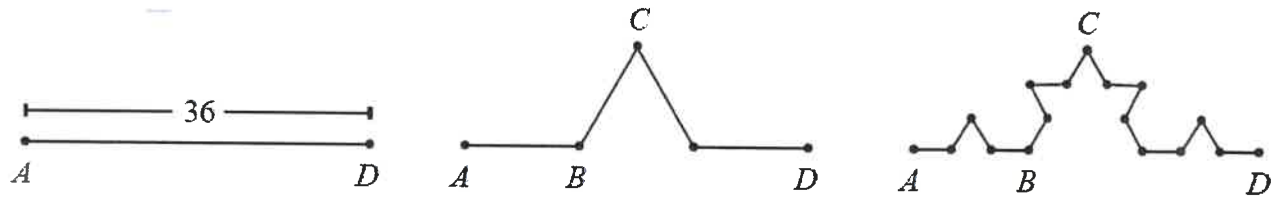
20. Lin starts with a line segment that has length 36 and adds a bump to it. She then adds bumps to each line segment of that path. The resulting figure is shown below on the right. All segments in each figure are of equal length. What is the total length of the figure on the right?



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For full answers please scan this code:

1. R1 220	2. 26	3. 2	4. 8	5. 36	5 marks each
6. 499	7. $\frac{18}{99}$ *or* $\frac{2}{11}$	8. 12:24	9. 24	10. 990	
11. 35	12. 5	13. 260	14. 47	15. 20	10 marks each
16. 288	17. 39%	18. 37	19. 12	20. 64	20 marks each